Plant fitness responses and insect-feeding choice in response to induced herbivory

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Introduction

Plants are constantly attacked by a large variety of herbivores, from small insects to large mammalian herbivores[1]. In response, plants use several defense mechanisms, such as compensatory regrowth of consumed biomass[2] and herbivore-deterring chemicals[3]. Increased foliage production from compensatory regrowth, however, provides more food for herbivores and increases the herbivore population size[4]. Additionally, the increase in resource allocation to compensatory regrowth may decrease resources available for vegetative spread[5] or increase plant susceptibility to insects by reducing resources for insect-deterring chemicals[6]. To understand these interactions between plants and herbivores, we quantified the plant fitness response and insect choice to plants exposed to cutting regimes that emulated mammalian grazing regimes.

Objectives and Predictions

- **Quantify plant fitness responses to herbivory:** Plants varieties are known to differ in their response to herbivory, specifically biomass production, but compensatory regrowth in one fitness component will may not portend compensatory regrowth in another fitness component.
- **Quantify insect choice to plant responses**: When insect larvae are given a choice between plants from the same variety that have been exposed to different cutting regimes, the insect larvae will prefer the plants with the highest compensatory regrowth.

Methods

- **To quantify the plant fitness response**, we compared two varieties of orchard grass, *Dactylis glomerata,* with known differences in compensatory regrowth, Athos and Olathe[7,8]. Athos will have high compensatory regrowth under high frequency cutting regimes, and Olathe will have high compensatory regrowth under low frequency cutting regimes.
- We emulated variation in herbivory frequency by cutting plants to 10 cm at one of three cutting frequencies[9]: 1) no cutting/herbivory, control, 2) cut every other month, bimonthly, and 3) cut every month, monthly. We had 18 plants of each variety in each treatment.
- To measure plant fitness, every month we recorded the stem count for each plant and the height of the longest leaf. When we imposed herbivory, we also dried and weighed the harvested material.
- We compared stem count, leaf weight, and biomass per stem between treatments and varieties using generalized linear mixed effects models with variety, date, and treatment as fixed effects and location in greenhouse as a random effect.
- **To quantify insect choice of grass plants from different cutting regimes**, we used plants from the plant fitness experiment and measured preference with a 4-way insect olfactometer (Figure 2).
- A total of 40, (20 for each grass plant variety) 17-day old *Spodoptera frugiperda* larvae were exposed to air for an hour from four sources: 1) no plant, 2) control plant, 3) plant damaged monthly, and 4) plant damaged every other month.
- The preferences of plants by the insects were quantified by a χ^2 test.

Figure 1. Experimental Grass plants in the greenhouse.

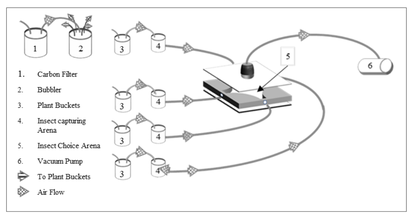

Figure 2. A schematic of the insect olfactometer used in the experiment. The insects were placed in the insect choice arena and given the choice to move to any of the 4 insect capturing areas. The different olfactometer parts are numbered in the order of the air flow.

Results and Discussion: Plant Fitness

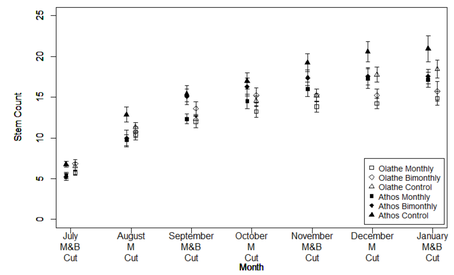

Figure 3. Stem count throughout the experiment for the two grass plant varieties (mean± SEM). M and B represent monthly and bimonthly cutting treatments. Contrary to expectations, varieties did not respond differently to the treatments (interaction between variety and treatment: χ^2 = 1.46, df = 2, p = 0.483), but there are general differences between the two varieties (χ^2 = 45.32, df = 6, p = <0.001) and three treatments (χ^2 = 28.84, df = 12, p = <0.001) through time.

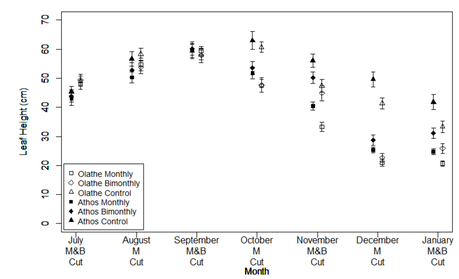

Figure 4. Leaf height versus time of the two grass plant varieties (mean± SEM). M and B describe monthly and bimonthly cutting treatments, respectively. Varieties did not differ in their response to the cutting treatments (χ^2 = 0.475, df = 2, p = 0.789), but leaf heights did differ between the treatments (χ^2 = 236.0, df = 12, p = <0.001) and between the varieties (χ^2 = 94.06, df = 6, p = <0.001), especially obvious in the months with less sunlight available. This suggests that the variety responses and treatment effects occur mainly when plants are limited by resource, in this case light.

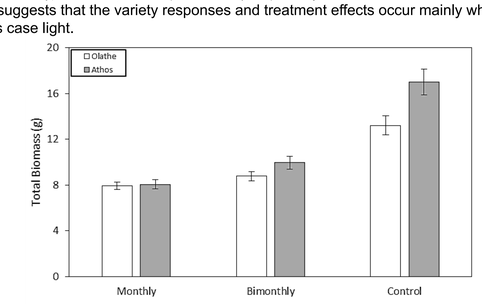

Figure 5. The total collected biomass in different treatments (mean± SEM). There was a significant difference in total biomass across varieties within treatments (χ^2 = 94.94, df = 2, p = < 0.001), where the control plants of both varieties produced the most biomass and the variety Athos produced more biomass in the control treatment and the bimonthly cutting treatment than Olathe produced.

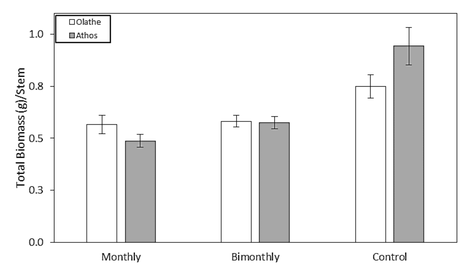

Figure 6. A graph showing the total biomass produced per stem in different treatments (mean± SEM). There was a significant difference in total biomass per stem produced across varieties within treatments (χ^2 = 0.356, df = 2, p = 0.0173). Again, control plants produced the most biomass per stem, but the two varieties differed in their biomass production, with Athos producing more biomass per stem in the control treatment but less biomass per stem in the monthly cutting treatment.

Results and Discussion: Insect Choice Experiment

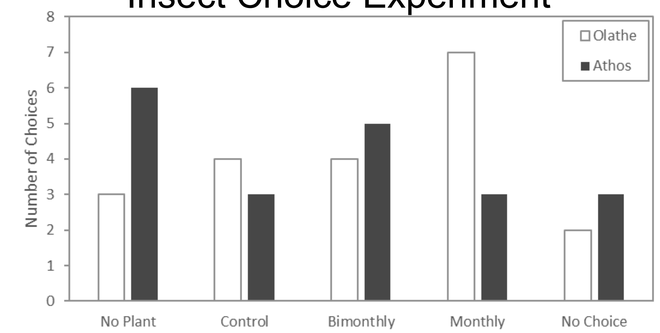

Figure 7. Insects did not appear to prefer one treatment over another treatment within a plant variety (χ^2 = 2.83, df = 3, p = 0.419). However, Olathe monthly and no plant in the Athos comparison was chosen more often than the other choices in the comparison. Larvae do not appear to move towards plants that offer more food via compensatory regrowth or away from more damaged plants.

Conclusions

Plant-fitness Responses

- There was no overall interaction between the different treatments and grass plant varieties (Figures 3 & 4).
- However, there were interactions between treatments and time and varieties and time (Figures 3 & 4). These fitness responses seem to be exemplified at different growing seasons.
- Contrary to our knowledge of growth in these varieties, Olathe did not produce more biomass in a monthly cutting regime, and Athos produced more biomass in a bimonthly cutting regime (Figure 5).
- We found that Athos had a higher fitness response than Olathe in a bimonthly cutting regime (Figure 5).

Insect Choice in Different Cutting Regimes

- There was no significant difference in insect choice of plants in different cutting regimes (Figure 9).
- Although we did not measure insect choice where larvae could choose between the varieties, insect preference tended to be greater for Olathe cut monthly than for Athos cut monthly, which could indicate that Athos produced more insect-deterring chemicals than Olathe does (Figure 9).

Acknowledgements

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References

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